Response to Editor and Reviewer Comments

Manuscript TTE-Reg-2026-09-3162: "Machine learning in battery management systems: a structured scoping review and classification of 103 peer-reviewed studies, 2019–2026"

Reworked manuscript: "Machine learning in battery management systems: a two-tier scoping review of 799 eligible studies with in-depth classification and validation audit"

We thank Professor Gao, the Associate Editors, and all four reviewers for a serious and, in the main, accurate set of critiques. The rejection identified real weaknesses: the corpus was curated rather than sampled while parts of the text read it as a field measurement; the validation-protocol taxonomy was under-defined and its arithmetic unreconciled; the classification rested on a single coder without a reliability statistic; the transportation-electrification fit was asserted rather than demonstrated; and the manuscript was not positioned against its closest prior review. Since receiving the decision we have substantially reworked the manuscript around those charges. This dossier responds point by point and is written to accompany a resubmission. Because the original submission was rejected rather than returned for revision, we state up front that the reworked manuscript differs from the version the reviewers saw in more than wording; the differences are listed in the next section and each is traceable to a specific reviewer comment.

A. What changed relative to the submitted version (disclosed in full)

- Retitled and redesigned as a two-tier scoping review. Tier 1 is the full 799-record rule-based eligible corpus (every eligible record included; no function quotas, year caps, or citation thresholds at this tier), from which bibliometric shares are measured with released rule-based function tagging and explicit capped-retrieval caveats. Tier 2 is the 103-study classified subset, whose quota-and-citation-ranked construction history is now disclosed in full and which is never generalized to the field. (R1.1, R2.1, R3-b1)
- The validation-protocol taxonomy is now defined before use, with two explicit stringency definitions (protocol-only and broader), an absorbing rule for borderline cases, and a released per-paper protocol audit (protocol_audit.csv) so every count reconciles row by row. (R1.2, R3-b6)
- A blind 30-paper re-coding reliability study was added, with per-field Cohen's κ and Krippendorff's α, an itemized disagreement table, locked and hash-tracked coder sheets, and a label-noise sensitivity analysis whose Monte Carlo noise model is now stated explicitly in the text. (R1.4, R2.2, R3-b4)
- A transportation-electrification cut of the corpus was added (fleet-tag share, fleet-data versus laboratory counts, venue concentration with numbers), together with a requirement-level mapping of BMS functions onto UN GTR 20, GB 38031-2020/2025, ISO 26262:2018, ISO/PAS 8800:2024, and ISO 21448. (R1.3, AE-1)
- The LLM discussion was corrected against its cited evidence, including the estimator role of the HA-LLM field-data study and the stationary-storage scope of the incident-analysis study, with a prior-art caveat. (R3-b7)
- A dedicated positioning against prior reviews was added, including a detailed comparison with the 2026 performance-centric EV-powertrain review of Prasanna et al. (Discover Artificial Intelligence 6:141), which two reviewers asked for. (R3-b8, R4)
- The complete evidence base is deposited at Zenodo (DOI 10.5281/zenodo.23436202) with a full artifact list; the AI-tool disclosure is consolidated into a single statement naming each tool and its purpose. (R1.4, R2 AI concern)
- The condensed submission form is 18 pages (ieee_submission/tex/mlbms.pdf, 105 references); an extended form (ML-BMS-Review.docx/pdf, 40 pp., 119 references, 13 tables) carries the same numbers with fuller discussion. All counts regenerate from the deposit.

B. Associate Editor and Area Editor Comments

AE-1. "As a review paper, this work does not cover sufficient directions of research or domains of applications. It also lacks sufficient outlook and insights into future research topics."

Coverage is extended along two axes. Horizontally, the Tier-1 eligible corpus measures the screened literature across all eight management functions (union-share bibliometrics over 799 records: SOH 20%, SOC 17%, RUL 13%, thermal 12%, fault/safety 12%, charging 8%, balancing 5% of records), so the review now reports domain coverage as a measurement rather than an assertion. Vertically, the validation audit, the standards mapping, and the per-function critical assessments add the analytical depth the AE found missing. On outlook: a new six-direction forward agenda (duty-cycle-aware validation; federated fleet evidence; benchmark diversification; pack- and vehicle-level ML aligned with UN GTR 20/GB 38031; functionally safe learning against ISO 26262/ISO/PAS 8800/SOTIF; honest foundation-model integration) replaces the earlier generic future-work paragraph (extended Section 9.8; condensed Section IX-C), and the reporting checklist is framed as an instrument that makes the next literature cycle auditable by construction.

AE-2. "The review manuscript does not demonstrate the level of technical depth, critical synthesis, and comprehensive state-of-the-art assessment expected for publication in an IEEE journal."

The rework adds the quantitative layer the AE called for: defined validation-protocol classes with per-paper assignments; reconciled counts (42 of 88 primary papers under strong protocols; 17 on information-leaking splits); a method-family contrast tested with Fisher's exact test, odds ratios, and 95% confidence intervals under two stringency definitions; and a label-noise sensitivity analysis with an explicit Monte Carlo model. The synthesis is now protocol-centric rather than narrative: each function chapter closes with a critical assessment of what the evidence actually supports (e.g., Sections 4.3, 5.6), and the safety-standards mapping converts the gap analysis into requirement-level statements an automotive reader can check. We accept that the submitted version fell short on these dimensions; we believe the reworked manuscript meets the bar, and we leave that judgment to the editors.

C. Reviewer 1

R1.1. The corpus is a curated reading list, not a systematic sample; every distribution in Section III is partly written in by the quota rule; either replace the quota step with a rule-based criterion applied to all 799 eligible records, or restate the Section III numbers as descriptive of this curated corpus and remove the field-level readings.

We did both, in a structured way. The 799 eligible records now constitute Tier 1 of the design and are all included, subject only to venue screening; the quota step survives only inside the Tier-2 in-depth subset, where its history is disclosed in full (stratified draw of 107 balancing functions and years; citation-ranked, deterministic within strata, walked deeper for recent years; 6 landmark seeds; −19 exclusions with reasons; +10 targeted additions; 103 = 98 pool + 5 seeds). Field-level claims now come only from Tier-1 measurements: function shares from rule-based title tagging with released term lists (validated against the subset's full-text labels, 71% agreement), and year counts presented as capped-retrieval counts with an explicit statement that they do not support absolute-growth claims. Tier-2 distributions are labeled as corpus composition in every section, and the abstract and conclusion no longer draw field-level readings from them. Section 2.3 states the within-stratum selection rule (deterministic citation ranking) that the reviewer correctly noted was missing.

R1.2. The central finding is not reproducible from the numbers as given; the strong-protocol category is never defined; per-protocol counts cannot be combined to yield the stated share; 10 versus 7 cross-chemistry studies are unrelated in the text; 15 reviews versus Table's 12; an audited table and a test on the contrast would settle this.

All four sub-points are fixed. (i) The taxonomy is now defined before use: four validation-protocol classes (within-cell split; unsupervised k-fold; cross-cell holdout with cross-chemistry and cross-domain extensions; online/real-vehicle validation), aggregated into strong (the isolation protocols) versus weak (the leaking splits), with environment-only categories (experimental hardware, simulation, case study) reported separately and never mixed into strong-protocol percentages; two stringency definitions are stated and every count says which it uses. (ii) The arithmetic now reconciles exactly and is auditable: 24 cross-cell + 7 cross-chemistry + 3 cross-domain + 8 online/fleet = 42 strong of 88 primary papers (48%); 12 within-cell + 5 k-fold = 17 weak; 29 environment-only (20 hardware, 9 simulation/case); per-paper assignments are released (protocol_audit.csv and the valid field of papers.csv) and were re-derived independently for 25 primary papers by the blind re-coding study. The 10-versus-7 relation is now stated: the 10 transfer validators comprise 7 cross-chemistry and 3 cross-domain, and the 7 are the studies behind the "validated across chemistries" count. (iii) The review count is reconciled as 12 method-family reviews plus 3 review-type syntheses coded under their dominant method family = 15 review-labelled papers, with the reconciliation carried in the terminology section and the function-by-year table’s Reviews column. (iv) The contrast is tested: under the protocol-only definition, physics-informed 8/22 (36%) versus others 34/66 (52%), Fisher exact p = 0.32, OR 0.54 (95% CI 0.21–1.46); under the broader definition, 13/22 (59%) versus 49/66 (74%), p = 0.19, OR 0.50 (95% CI 0.19–1.36); neither is statistically significant, both are reported, and the text claims neither as a model-family effect.

R1.3. The scope is not yet transportation electrification; the standards appear as motivation and never return; a mapping from BMS function to safety-standard requirement would belong here.

The rework adds exactly that mapping, as a table and as an inference rule. Each BMS function is mapped to what UN GTR 20 (including the Amendment 1 thermal-propagation provisions), GB 38031-2020/2025, ISO 26262:2018 with ISO/PAS 8800:2024, and ISO 21448 demand of a deployed function, alongside the corresponding subset status (e.g., 3 of 13 SOC primary papers under a strong isolation protocol; 0 of 8 charging and 0 of 7 thermal primary papers; 2 of 7 fault/safety papers online on fleets). The section states its inference rule explicitly: within-cell and k-fold evidence cannot support a standards argument because the safety case is about unseen cells and unseen duty. The corpus itself is now cut for this readership: about 2% of eligible records carry fleet/cloud tags; 10 of 88 primary papers use real vehicle or fleet data against 36 on laboratory cells; venue concentration is quantified (Journal of Energy Storage 103 of 799 eligible records; IEEE-prefixed venues 9.0%); and the forward agenda names pack- and vehicle-level ML as a first-class direction.

R1.4. DOI resolution verifies existence, not metadata; reference [10] is attributed to an RFID journal; the artifact link is a GitHub account rather than a DOI-backed deposit; classification was one coder with a corrective second pass; a blind double-coding with an agreement statistic would cost little.

Each item is addressed. (i) The verification claim is now scoped to what DOI resolution does, in the limitations text. (ii) The RFID-venue record is flagged in the manuscript itself rather than silently dropped, and the released audit file (final_audit.json) carries each reference’s Crossref-resolved title and venue at retrieval time so such artifacts are visible rather than hidden. (iii) The artifact of record is now a Zenodo deposit, DOI 10.5281/zenodo.23436202, containing the classified dataset, the eligible pool, the record-level screening log, the protocol audit, the codebook, the reliability files, the conference set, the sensitivity script, and all generation scripts; the GitHub repository is described as a development mirror, which is what item 12 of our own checklist asks of others. (iv) The blind double-coding the reviewer proposed was run: two co-author coders not involved in the primary coding re-coded a 30-paper stratified random sample under the released codebook, blind to the released labels, to the population statistics, and to each other, with sheets locked and hash-tracked at completion. Agreement is strong on three of five fields (function $\kappa = 0.92$; method family $\kappa =$

0.88; data source κ = 0.88) and below threshold on chemistry (κ = 0.55) and validation protocol (κ = 0.48), with the evidence-access constraint disclosed (full-text access to 7 and 2 of the 30 papers). Because the coders are authors of the study, the manuscript states plainly that the blinding was procedural rather than external, and all headline validation counts carry a sensitivity analysis built from the observed coder-versus-label disagreement.

D. Reviewer 2

R2.1. Function quotas, year balancing, targeted additions, and quota-based exclusions shaped the final 103 papers; publication trends, application shares, and method-family growth cannot be interpreted as field-wide trends.

Agreed, and the rework enforces it structurally. Trends and shares are now either measured on the quota-free 799-record Tier-1 pool (function union shares; capped-retrieval year counts) or explicitly labeled as within-subset patterns. The abstract, Section 3, and the conclusion were rewritten so that no field-level reading rests on the curated subset; e.g., the method-family evolution paragraph states the primary-paper counts year by year with the partial-2026 caveat and calls the trajectory a within-subset observation, and the conclusion phrases the field-level statement only where Tier-1 measurement supports it.

R2.2. One author coded all papers; the co-author review was corrective rather than independent; without independent double coding and an agreement measure it is unclear how much of the pattern reflects coder decisions.

The reliability study now exists and is reported without embellishment: 30-paper stratified random sample (seed-pinned, reproducible), released codebook with a decision tree and an absorbing rule, two co-author coders blind to the released labels and to each other, per-field κ and α, an itemized disagreement table, and locked hash-tracked sheets. The manuscript states the limits honestly: the coders are authors (procedural, not external, blinding), and their full-text access was uneven, which the two low-κ fields (chemistry, validation protocol) plausibly reflect. Quantitatively, all headline validation counts now carry a label-noise sensitivity analysis: the strong-protocol share spans 40–59% under plausible label error, and no qualitative conclusion depends on where in that range the truth lies. The noise model is stated in the text (4,000 seeded Monte Carlo draws; per-draw flip rate ~ Normal, centered on the observed coder-versus-label bucket error rate of 0.35, SD 0.02, truncated at zero) and the script is released.

R2.3. The paper does not demonstrate that within-cell or k-fold validation overstates accuracy; it identifies a methodological risk, not direct evidence of inflated accuracy.

The reviewer’s reading is the one we now adopt verbatim in substance: the review claims a methodological risk, not measured inflation. The text was purged of causal accuracy claims: within-cell and k-fold splits are said to leak trajectory information and to produce errors that do not survive cell-to-cell variability — a statement about protocol design — and the conclusion phrases the finding as the proliferation of published accuracy claims having outpaced the validation evidence behind them, not as a demonstration that reported accuracies are inflated. We also explain, in Section 5.6, why cross-paper pooling of reported errors is not defensible for this literature (metric heterogeneity), which is precisely why a between-group accuracy comparison was not attempted.

R2.4. The physics-informed claim (13/24 versus 49/67) has no confidence interval, significance test, or adjustment; it may reflect corpus composition; it should be treated as exploratory.

Fixed as specified. Both stringency definitions are reported with Fisher’s exact test, odds ratios, and 95% confidence intervals (protocol-only: 8/22 = 36% versus 34/66 = 52%, p = 0.32, OR 0.54, CI 0.21–1.46;

broader: 13/22 = 59% versus 49/66 = 74%, p = 0.19, OR 0.50, CI 0.19–1.36); the text states that neither difference is significant, that confounding is unadjusted (physics-informed work skews early-stage and simulation-heavy), that n = 22 is small, and that the result is an untested hypothesis rather than a settled finding — the same words appear in the conclusion. The label-noise sensitivity analysis leaves the contrast non-significant in 95% of draws.

R2.5. The conference sensitivity analysis compares abstract-level evidence with fuller journal assessment; a missing validation statement in an abstract does not mean validation was absent; the set is small and citation-ranked.

The conference section is reframed as what it can be: a descriptive check of stated-evidence transparency, not a test of venue-selection bias. The comparison is now between abstract-stated validation on the conference path and abstract-stated experimental-hardware validation among journal papers' indexed records, with the truncation caveat the reviewer raised stated in the text, the pre-stated mechanical inclusion rule applied to all 14 retrieved papers (the conventional model-based estimators the reviewer noted are excluded from the learning-path comparison by that rule and retained for transparency), and the set released in full. The section's conclusion is now about reporting transparency, not about conference practice, and it feeds the checklist rather than any venue claim.

E. Reviewer 3

R3-b1. Function-quota selection introduces selection bias; proportions cannot establish which functions or methods dominate the literature; claims should be restricted to the selected corpus.

Restriction is now structural: the Tier-2 subset is never generalized, and where dominance claims are made they are Tier-1 measurements (rule-based tagging over all 799 eligible records, term lists released, tagger validated at 71% against the subset's full-text labels). The two levels are labeled everywhere they appear, including figure captions.

R3-b2. The selection process is difficult to examine; 107 candidates, 19 exclusions, and 15 additions reach 103 but do not explain which studies were removed for ineligibility versus balance; a record-level screening table with reasons would make selection verifiable.

The record-level table exists and is released: screening_log.csv carries all 1,550 screening decisions with outcomes and reasons, the selection files identify every inclusion and drop at every stage with the five forced seeds annotated, and Section 2.3 of the manuscript now narrates the funnel with the categories of exclusion (corrigenda, misindexed records, scope mismatches, quota-driven trimming) distinguishable. The flow figure matches the narration, including the 98 + 5 decomposition.

R3-b3. Section II-B admits reviews only to the cross-cutting cluster, yet Table II assigns reviews to several categories; Table I reports 12 reviews while Section III-C mentions 15.

The definitions are reconciled and stated once, then used consistently: 12 method-family reviews are admitted under the review cap, and 3 further review-type syntheses are coded under their dominant method family, giving 15 review-labelled papers; the function-by-year table carries a Reviews column that states this reconciliation per function, and the terminology section defines primary-application (88) versus review-labelled (15) papers. Where the earlier text said reviews sit only in the cross-cutting cluster, it now says reviews are admitted to the cross-cutting cluster and that review-type syntheses are coded under their dominant family.

R3-b4. The coding process lacks an independent reliability check; a second independent coding of a subset with a codebook and dispute rules would establish reproducibility.

Done as proposed, with the honesty limits stated: 30-paper stratified random sample, released codebook with decision tree and absorbing rule, two co-author coders blind to the released labels and to each other, locked hash-tracked sheets, per-field κ/α, and an itemized disagreement table (see the response to R2.2). The manuscript does not claim external independence it does not have; it quantifies what procedural blinding yields and attaches sensitivity ranges to every headline count the coding supports.

R3-b5. Method-evolution analysis is weakened by uneven sampling across years (one 2019 landmark, seven 2026 papers, searches ending early September); Figure 3 displays percentages despite small, differently selected denominators; displacement claims should be expressed as within-sample patterns.

The evolution figure now displays paper counts, not percentages; the 2019 seed is shown as what it is (a forced landmark); 2026 is labeled a partial year at the September retrieval date; and the physics-informed trajectory is stated as primary-paper counts (1, 2, 3, 3, 4, 4, 5 across 2020–2026) with the monotonicity remark explicitly confined to the within-subset primary trajectory. The measured-tier year counts (Tier 1) are reported separately, with the 2024 retrieval hole documented, and the text states that neither level supports absolute-growth claims.

R3-b6. Cross-cell holdout, cross-domain transfer, laboratory hardware, and fleet operation test different properties; coding rules for borderline cases are unclear; the classification underlying the central comparison should specify thresholds, study-level assignments, and distinguish generalization from deployment evidence.

The codebook now specifies each threshold and the absorbing rule (a study is credited with the strongest protocol it reports for its headline claim), with a decision tree and worked examples; study-level assignments are released (protocol_audit.csv); and generalization evidence (isolation protocols) is distinguished from deployment evidence (operating environment) by construction — the environment-only categories are reported separately and never mixed into strong-protocol percentages. The fleet table additionally separates data use (10 papers on real vehicle/fleet data) from online validation (8 papers), which are different measures, and cross-tabulates them.

R3-b7. The LLM discussion does not match its evidence: reference [15] is an LLM-based SOH estimator on field data with uncertainty-aware controls, not merely an orchestration layer; reference [14] concerns stationary-storage incidents, not onboard EV control; Table V should distinguish what the studies demonstrate from the authors' proposed restrictions.

Correct on both counts, and the section is rewritten accordingly. The HA-LLM study is now described as what it is: an LLM-based SOH estimator that aligns resampled charging-current sequences with the LLM embedding space, injects degradation priors by prefix adaptation, and wraps the model in uncertainty-aware Monte Carlo gating with substitution by a physics-consistent estimator, validated on one year of field data from 100 production vehicles with cross-vehicle few-shot and zero-shot evaluation — i.e., it already occupies the estimator seat, under exactly two of the guardrails we propose. The incident-analysis study is described as stationary, station-side technology, not onboard control. A prior-art caveat was added (LLM-based battery estimation exists in preprints, conferences, and a 2025 journal letter outside the screened venues), and Table 9 now states explicitly which column reports what the studies demonstrate and which column is our normative guidance.

R3-b8. A 2026 systematic review has already classified 95 studies across battery-management applications and computational methods; earlier reviews examined similar approaches; the contributions are not clearly distinguished; a more direct comparison is needed.

A positioning section now does this directly (extended Sections 1.3 and 9.7 with a comparison table; condensed Sections I-B and IX-C with Table VII). The closest recent cross-functional review — the

performance-centric review of Prasanna et al., Discover Artificial Intelligence 6:141 (2026), covering 240 publications across the EV powertrain — is compared in detail: in scope (BMS functions in depth versus powertrain breadth), in unit of analysis (validation-protocol audit versus aggregation of reported RMSE/R^2), and in question asked (which claims can be checked, versus which algorithm family performs best as reported). The comparison is framed as complementary rather than adversarial: a performance-centric aggregation inherits, unmeasured, exactly the validation weaknesses this review quantifies. The same section positions the work against the function-specific and method-specific reviews cited in the introduction, and the contribution list now states the delta explicitly.

F. Reviewer 4

R4. "A thorough review, however, it remains incremental in nature. Numerous reviews are available, to cite one: Prasanna et al. (2026), Discover Artificial Intelligence, 6(1), 141."

The detailed response to R3-b8 applies, and the comparison with Prasanna et al. is now in the manuscript in the reviewers' requested form: a dedicated paragraph plus a per-axis table. In one sentence, the delta is: prior reviews, including the performance-centric powertrain review, synthesize what is claimed and how well it is reported to perform; this review additionally measures how those claims are validated (protocol classes, per-paper assignments, method-family contrast with significance tests and sensitivity ranges), ties the gaps to three automotive safety standards at requirement level, and releases the record-level dataset, screening provenance, codebook, and re-runnable pipeline that make every count auditable. We believe that combination — not any single finding — is the contribution, and it is the one no available review offers.

G. Reviewer 2's assessment that the manuscript gave "the strong impression that it was generated almost entirely with AI"

We take this concern seriously and answer it with disclosure and evidence rather than assurance. The manuscript now carries a single consolidated declaration naming each tool and its purpose: an AI language model (GLM, Z.ai) assisted in drafting and adapting the literature-pooling, analysis, and figure-generation scripts; ChatGPT, Grammarly, and QuillBot were used for language editing. The declaration also states what no AI tool did: no AI tool decided study inclusion, and no AI tool assigned or altered any classification label — all screening decisions and all released labels were made by the authors, and the released reliability files (locked coder sheets, blinding memo, disagreement tables) document the human coding evidence. Every reference was verified by direct DOI resolution, every count and figure regenerates from the deposited dataset, and the authors take full responsibility for the content. We hope the reworked methodological substance — defined taxonomies, tested contrasts, disclosed limitations — also addresses the underlying judgment behind the impression.

H. Summary

The rejection identified that the manuscript asked the reader to trust numbers the design could not support. The rework fixes the design rather than the wording: a two-tier evidence base that separates measurement from curation, a validation taxonomy with audited per-paper assignments and tested contrasts, a reliability study with its limits stated, a transportation-electrification cut with a safety-standards mapping, a corrected LLM discussion, an explicit positioning against prior reviews including the one two reviewers named, and a DOI-backed deposit behind every number. We believe the reworked manuscript is a different and substantially stronger paper, and we would be glad to submit it to the Transactions as a new manuscript if the editors consider that appropriate; otherwise this dossier accompanies its submission elsewhere.

I. Post-rework quality audit (second pass)

A further internal audit of the reworked manuscripts surfaced eight residual issues. All are corrected in both the extended and the condensed versions, and the corrected artifacts are re-released:

- Prasanna et al. corpus size stated precisely. The comparison paragraph now reads "240 peer-reviewed publications (included from 361 initially retrieved records under its PRISMA screen)". The published abstract's funnel is 361 records retrieved, 33 duplicates removed, 88 excluded at screening, 240 included; the figure of 95 studies quoted in one review does not appear in the published abstract, and the count is now stated so it can be checked against it directly.

- The §8.4 "field's first LLM estimator study" claim corrected. It now reads "the subset's earliest LLM estimator study." The earlier prior art — Poh et al., IEEE Transactions on Energy Conversion, 2025 — is disclosed as a recall miss: that record appears nowhere in the released 1,550-record retrieval, so it is a limitation of the query design, not a venue exclusion. Tables 8, 9, the condensed version's LLM section, and its §9.5 counterpart now use the same framing.

- Tier-1 tagging described once, accurately, with its limits disclosed. Tags are inherited from the retrieving queries (a record carries every area whose query returned it); the previous "keyword pass over titles" wording described the relevance gate, not the tagger, and has been removed from all four places. The disclosure now states that 24 of 799 eligible records carry only the generic "Other" tag, that 202 carry no core-function tag, and that the validation against full-text labels is a lenient 71% (53 of 75 core-area papers, any-tag match).

- Evidence-strength corrections in the validation section. "Optimistic errors" claims are hedged to "can produce optimistic errors" (§3.4, Table 5, §9.1); the physics-informed "rising" claims are scoped to the classified subset, with the eligible-pool numbers now shown (1, 10, 12, 18, 2, 3, 9 tagged records in 2020–2026; shares 1.8–11.8–8.6–8.1–3.6–2.4–7.9% — non-monotonic); and Wilson 95% confidence intervals are reported on the two headline rates — 42/88 strong (47.7%, CI 37.6–58.0%) and 8/88 online (9.1%, CI 4.7–16.9%) — matching the §1.4 promise.

- Comparison-paragraph overreach hedged. The Prasanna aggregation is now said to "may inherit" the validation weaknesses quantified within this review's own subset (the 17-of-88 rate is this sample's, not theirs), and Table 12's claims about prior reviews are hedged ("no released record-level dataset that we could locate"; "none of them, to our knowledge").

- Standards now carry real citations with document symbols. Six standards documents were added to the reference list: UN GTR No. 20 (ECE/TRANS/180/Add.20, established in the Global Registry 13 June 2018); its Amendment 1 (UNECE EVS-30-C01/Rev.4, 11 September 2025, thermal propagation with two compliance options, listed for the WP.29 session of 17–20 November 2026); GB 38031-2025 (published 28 March 2025; effective 1 July 2026 for new type approvals and 1 July 2027 for existing models); ISO 26262:2018; ISO/PAS 8800:2024 (published December 2024); and ISO 21448:2022. The previous "inspired by ISO 26262 and ISO 6469-1" attribution, which we could not verify against the proposal text, was replaced with the proposal's own description of the two compliance options.

- Cross-reference and journal-targeting fixes. §3.1 now cites Fig. 4 (left panel) for the eligible-corpus counts, and Fig. 4's caption describes both panels; "this journal" was replaced with neutral wording; and the condensed version's abstract-level readership phrasing was left as venue-neutral.

- Sensitivity analysis reframed as a scenario envelope, with a directional scenario. The label-noise analysis is now introduced as a scenario envelope rather than an error model, with the caveat that symmetric random flipping attenuates contrasts toward the null. Because 16 of the 19 observed coder-label

disagreements have the coder defaulting to weak, a strong-to-weak-only scenario was added: it spans 22–35 of 88 (25.0–39.8%), with the family contrast non-significant in 96.3% of draws (symmetric envelope unchanged: 35–52, 39.8–59.1%, 94.4%). The released script now reproduces its output from the released gold file (observed coder-vs-label rates 0.37 and 0.27; the previously shipped 0.40/0.30 came from a pre-recode gold version), and the conference-venue check is explicitly labelled a descriptive, abstract-level comparison that feeds no headline statistic.

None of these corrections changes a headline conclusion; every changed number was re-derived from the released data files and both manuscripts were rebuilt end to end.

J. Third-pass audit: retrieval recall, subset scoping, and statistical honesty

A third audit round examined the reworked manuscripts' retrieval assumptions, claim scoping, and statistical phrasing. It produced two new released analyses (a recall audit and a minimum-detectable-effect analysis), two abstract-level probes, a terminology-and-reconciliation box, an interpretation-rules box, and wording corrections throughout both versions:

M2 — Retrieval recall is now quantified, not assumed.

A recall audit (released as recall_audit.py with cached Crossref responses and recall_audit.json outputs) matches the deposited reference lists of the 15 included reviews and 6 seeds against the 1,550-record retrieval. Of 2,508 references, 1,367 are pre-2020 (outside the window by design); of the in-scope-looking, DOI-bearing, 2020-onward references (649 with multiplicity, 586 unique), the retrieval contains 112 (17.3% per-source; 96 of 586 unique, 16.4%). The audit states its own limits (citation-conditioned denominator, title-level gate, DOI-dependent matching). Per-query yields are released with it and show every query family is design-capped (top 20 per pass, 5 passes, 100 records per family; 2,973 query-pass records deduplicating to 1,550) — yields are caps, not saturation. The missed TEC 2025 GPT-4 estimator is cited by none of the included reviews either, so reference-list searching would not have found it. "First/earliest" wording is now subset-scoped in the abstract, §3.2, Table 11, and Fig. 8 of the extended version, and in the condensed abstract, II-B, and its standards table.

M3 — Every headline is subset-scoped; representativeness probed, honestly.

The abstract now reads "Three structural weaknesses recur across the classified subset" and the conclusion "the subset's published accuracy claims"; a new Box 1 states five interpretation rules that every quantitative claim invokes. The suggested field estimate was attempted as a seed-pinned, 40-record uniform sample of the 799-record pool, coded at abstract level (abstract_probe.py/.json with released codes): 23 of 40 abstracts are publicly retrievable (Elsevier withholds the rest), and none of the 18 primary-looking pool records states an isolation or deployment protocol in its abstract, versus 5 of 38 abstract-screenable subset papers (Fisher $p = 0.16$). The honest conclusion is reported: indexed abstracts almost never carry protocol information in either tier, so the 48% strong-protocol share cannot be promoted from a curated-subset statistic to a field estimate without full-text access — and the scarcity of protocol statements in abstracts is itself folded into the case for the reporting checklist.

M4 — Non-significance is no longer phrased as quasi-equivalence.

A minimum-detectable-effect analysis (power_analysis.py/.json; analytic two-proportion power validated against Fisher-exact Monte Carlo) shows that with 22 versus 66 papers and a 51.5% baseline rate, 80% power at $\alpha = 0.05$ requires a gap of roughly 30–35 percentage points (odds ratio ≤ 0.26 or ≥ 4.0); the observed 15-point gap is detectable with only about 20–25% power, and the arms' Wilson intervals overlap almost completely (19.7–57.0% versus 39.7–63.2%). The abstract, §3.4, and the conclusion of the extended version, and the

abstract, III, and the conclusion of the condensed version, now say exactly that: no detectable difference, meaningful differences in either direction cannot be excluded.

M5 — Tier-1 function shares carry their caveat into the abstract.

The abstract's SOH/SOC/prognostics shares are now tagged as "query-inherited tag shares, read for ordering, not exact rates," matching the §2.3 disclosure.

M6 — The baseline question is now coded, at the level the evidence allows.

A seventh field — a baseline-comparison screen with values none / mentioned / stated same-data comparison — is added to §2.5 and the released codebook (which also defines the full-text rubric for future work), targeting exactly the question the checklist's item 6 poses. Results (§3.4): of 38 abstract-screenable primary papers, 32 name no model-based or classical baseline, 2 name one without stating the comparison, and 4 state a same-data quantitative comparison (versus an SVR, a 0.5C CCCV protocol, an equivalent-circuit/electrochemical model, and a rule-based controller). The bounds are stated with the numbers: abstract-level codes are lower bounds on comparison practice and exact measures of comparison visibility, and 49 of 88 abstracts (Elsevier-withheld) cannot be screened from public indexes at all.

M7 — Review-rigor deviations given PRISMA-ScR-level prominence.

A dedicated paragraph in §2.6 discloses: no protocol registration (PROSPERO does not register scoping reviews; no alternative registry was used); single-reviewer screening, qualified precisely — the title gate and venue screen are deterministic applications of released term lists with all 1,555 per-record decisions logged, but no second human screened them, the judgment-bearing stages were single-reviewer, and the only dual-coded element is the co-author-coded reliability study; and no per-record time-on-task logs (retrieval ran 4–5 September 2026; the rule-based screens execute in seconds; the judgment stages were not time-stamped, so effort cannot be audited). Tool assistance is bounded as the AI declaration states — no tool-assisted triage or ranking participated in screening; the one ranking used (citation ordering within strata) is deterministic and released.

M8 — The 799 are decomposed exactly; the taxonomy is stated once.

The 799 eligible records comprise 798 dated 2020–2026 (55/85/139/223/56/126/114) plus one early-access record Crossref-stamped 2027; the 2019 Severson seed is NOT among them — it is one of the five seed studies added individually to the 103-study classified subset (98 pool + 5 seeds). The abstract, Table 1's period criterion, and §2.2 now state this composition exactly. A new Box 2 carries the terminology (eligible corpus, classified subset, primary, review-labelled, family-coded), the full funnel (3,000 → 2,973 → 1,550 → 883 → 799 → 98+5 = 103 = 88+15), and the glossary, and the §2.5 terminology sentence now points to it instead of restating the split. §3.2's foundation-model sentence and Table 11 are subset-scoped, and Table 10's caption now names the stringency definition behind every "strong protocol" count.

M9 — The conference check stays descriptive.

The section's closing sentence no longer claims the journal corpus "does not understate" anything; it states the observed pattern (stated validation evidence no more visible in the eight conference abstracts than in the journal corpus) and says explicitly that eight truncated abstracts cannot carry an inference about venue populations.

M10 — Standards wording aligned and date-stamped.

GB 38031-2025 is now worded identically in §9.4 and Table 10 (and the condensed table): it requires a thermal-propagation warning ahead of hazard and no fire or explosion of the pack beyond the triggered cell, superseding GB 38031-2020's five-minute pre-warning criterion, with bottom-impact and 300-cycle fast-

charge tests added. GTR 20 Amendment 1 is date-stamped in both places — document symbol EVS-30-C01/Rev.4, and status "pending before WP.29 with adoption listed for its 17–20 November 2026 session, as of the September 2026 retrieval date."

Minor points — all addressed.

The abstract was de-nested and its asides folded into main clauses; the physics-informed result and the two-tier caveat are consolidated into Box 1 (with the §2.6 duplicate trimmed to a pointer); the 2024 "hole" is clarified as an observed count inside the capped retrieval, not an estimated missing volume (both versions); the Wilson intervals now carry a curated-subset caveat; the garbled Crossref author metadata for one reference (10.1016/j.jechem.2023.03.026 — given/family transposed, "Store" for "Stroe") is normalized to the published names (Vilsen, S. B.; Stroe, D. I.) with a registry-error note in the data-availability statement; the §3.3 balancing claim is softened to a single contrarian study; Table 8's gloss format is uniform again; Fig. 8's caption warns that milestone dating is citation- and retrieval-conditioned; the absorbing rule is defined where it is used (Table 6 caption and Box 2); union-share, environment-only, and protocol-only are defined in Box 2's glossary; Fig. 5's caption states n = 103; and the Zenodo deposit's resource type is corrected to dataset in the released metadata file (the published record needs a one-click metadata edit in the authors' Zenodo account, noted in the upload steps).

All new numbers regenerate from released artifacts (recall_audit.py/.json with cached Crossref responses; power_analysis.py/.json; abstract_probe.py, abstract_probe_raw.json, abstract_probe_codes.py, abstract_probe.json; the extended codebook). Both manuscripts were rebuilt and re-verified: the consistency checker re-derives 42/88 strong, 8/22 versus 34/66 ($p = 0.324$), 13/22 versus 49/66 ($p = 0.189$), 12+3 = 15 review-labelled, and 103 = 88+5+15−15 reconciliations; text sweeps confirm every stale phrasing listed above is absent from both PDFs.

K. Fourth-pass QA: the strong-to-weak bound, claim precision, and standards citations

A fourth audit round re-examined the sensitivity design, several claim-wording issues, the corpus/subset arithmetic, and the standards citations. All changes were recomputed from the released data where numbers were involved, and both manuscript forms were updated in parallel:

1 — The strong-to-weak scenario now uses the strong-label-specific rate, and is presented as a worst-case bound.

The previous directional scenario applied the coders' overall bucket error rate (0.37/0.27, mean 0.32) to strong labels only. That was wrong: the observed disagreements are concentrated on strong labels. Recomputed from the released gold_labels_30.csv and the two locked coder sheets, 13 of the 30 sampled papers are strong-labelled and each coder recoded 8 of those 13 weak (16 of 19 pooled disagreements are strong-label/weak-recode), a strong-label-specific rate of 0.62. The scenario now draws its flip rate from that rate (same 0.02 standard deviation, same seed and 4,000 draws): the strong-protocol count falls to about 16 of 88 (≈18%) at the observed rate, with a scenario envelope of 10–23 (11.4–26.1%), and the family contrast remains non-significant in 97.6% of draws. Because the coders worked from abstracts under a declared weak-if-unreported rule, the text now presents the scenario explicitly as a worst-case bound, not an estimate. The abstract bracket moves from "25% and 59%" to "11% and 59%", and its wording now reads "symmetric and strong-to-weak" (there is no weak-to-strong-only scenario). The released script (sensitivity_label_noise.py) and its JSON output were updated to compute and document both rates.

2 — The blind re-coding is no longer described as re-deriving the primary-paper assignments.

Section 3.4 previously said the 30-paper blind re-coding "re-derived 25 of these primary-paper assignments independently." The 25 of 30 is inter-coder agreement between the two blind coders on a sample that includes

review-labelled papers, not agreement with the released labels (their bucket kappas against the released labels are 0.22 and 0.42, §2.5). The sentence now states exactly that: the re-coding is an abstract-level inter-coder agreement check on a sample that includes review-labelled papers, not an independent re-derivation of the assignments. The condensed version carries the same correction.

3 — The 799/103 relationship is stated exactly, in every location.

The abstracts already described the 799 as 798 records dated 2020–2026 plus one early-access record stamped 2027 (the 2019 seed sits in the 103, not the pool). Section 2.3 nonetheless opened Tier 2 with "a 103-study subset of the eligible corpus," which is arithmetically wrong for the 5 individually retrieved seeds; it now reads "a classified subset of 103 studies — 98 drawn from the eligible pool plus 5 seed studies retrieved and verified individually," with the same fix in the condensed Section II-A and both cover letters. One location outside the manuscripts still needs attention: the published Zenodo record's description still reads "799 peer-reviewed journal articles (2019-2026)"; the corrected local metadata is in the deposit package, and ZENODO_UPLOAD_STEPS.md now flags the live-record edit explicitly.

4 — The abstract no longer gives tag shares that §2.3 disclaims; the union is now computed and reported.

The exact shares (20%, 17%, 13%) were dropped from both abstracts, which now name the three estimation functions and read the tags for ordering only. The previously uncomputed union behind "about half of eligible records" is now computed and reported where the per-tag baselines are given: the three estimation functions together tag 366 of the 799 records (45.8%, §3.3 extended; 46% in the condensed III-B, which already carried it), and both conclusions cite the number.

5 — The conference-venue check no longer draws the conclusion it just disclaimed.

Section 3.5 now reports that the comparison finds no evidence of a venue gap in stated validation evidence and, with eight abstracts on one side, cannot rule one out. The 25% vs. 23% percentages are dropped (raw counts kept), the "if anything, lower on the conference path" sentence is removed, and the section closes on the open question plus the reporting-standard implication for both venue types.

6 — The conclusion now leads with the validation audit.

The three conclusions are reordered: the claims-versus-evidence finding (previously third) now leads; the estimation center of gravity is second; the method-family sequence is third and explicitly scoped as a within-subset sequence with no measurable field-level rise. "Defining meta-finding" is removed from §3.2 (and "the defining finding" from the condensed §III), which now opens the paragraph with the observation itself.

7 — Standards citations now point at the formal ECE documents, and the GB 38031-2025 wording is verified.

The GTR 20 Amendment 1 citation now uses the formal working-party document ECE/TRANS/WP.29/GRSP/2026/4 (Proposal for Amendment 1 to UN GTR No. 20, submitted by the European Commission on behalf of the Informal Working Group on Electric Vehicle Safety), with the status report ECE/TRANS/WP.29/GRSP/2026/3 cited alongside it; the informal EVS-30-C01/Rev.4 symbol is gone from both manuscripts and the reference data. The status line is date-stamped to this finalization (18 October 2026): the proposal was considered at the 79th GRSP session (June 2026) and stands pending before WP.29 with adoption listed for the 17–20 November 2026 session; because the vote may have happened by the time of submission, the status line is on the pre-submission checklist. The GB 38031-2025 thermal-propagation wording in §9.4 and Table 10 was verified against the standard's text as reported by the standards bodies and official Chinese coverage: once single-cell thermal runaway is triggered, the pack or system must neither fire nor explode throughout a two-hour observation window, a thermal-event alarm must still be issued, and smoke

must not harm occupants — superseding GB 38031-2020's criterion of an alarm at least five minutes before any fire or explosion. The previous "warning ahead of hazard" phrasing (which described the superseded 2020 logic) is corrected in both forms.

8 — Smaller items.

Table 11 now reads "First two LLM-based studies appear 2025–2026 in the subset." Reference [10] (the IEEE Journal of Radio Frequency Identification SOC study) now carries a one-line disclosure note in both reference lists: a lithium-ion SOC estimation study published in an identification-technology venue, retained on DOI-verified method content. Section 3.1 states explicitly that no query pass was dedicated to January–May 2024 (the fetch filters put the recency pass at 1 June 2024 and end the mid-window pass in December 2023, so the hole is January–May). Year balancing is described as partial wherever it is invoked, with the subset's per-year span of 7–22 papers stated in §3.1 (and §2.3/"partly balancing publication years," plus the Fig. 3 caption). A formatting defect was also fixed: the extended abstract's emphasis markers now render as italics instead of literal asterisks.

L. Fifth-pass QA: internal inconsistencies and presentation

A fifth audit round fixed seven internal inconsistencies (four of them arithmetic or corpus-data mismatches) and restructured the presentation of robustness material. Every corrected count was re-derived from the released data before editing:

1 — Screening-log row count harmonized (1,555 vs. 1,550).

The released screening_log.csv carries 1,555 rows: the 1,550 retrieved records plus the five individually retrieved seeds. Section 2.6 previously said "1,555 per-record decisions" while Section 2.3 and the Fig. 2 caption said "all 1,550 screening decisions." All three locations (and the two parallel spots in the short version) now use the same phrasing: the log carries 1,555 rows — the 1,550 retrieved records plus the five individually retrieved seeds.

2 — Deduplication arithmetic corrected (3,000 / 2,973 / 1,423 / 1,550).

The first retrieval paragraph said "3,000 retrieved records; after DOI-based deduplication (1,450 duplicates removed) 1,550 unique records remained," but the funnel (Box 2, Fig. 2) has 2,973 records entering dedup. Both cannot be right: 3,000 − 1,450 = 1,450 counts the 27 rows dropped by the document-type, year, and title-completeness filters as if they were duplicates. The text now states the funnel exactly as the data does: 3,000 fetched rows; 2,973 after filters; 1,423 duplicates removed; 1,550 unique records. The short version was fixed in parallel.

3 — Review-admission claim aligned with Table 3.

Section 2.4 said review-only papers were "admitted only into the cross-cutting cluster, capped at 12," but the released labels put only 5 of the 15 review-labelled papers in the cross-cutting cluster (the other 10 review the functions they survey: 4 state-estimation, 2 fault-and-safety, 2 prognostics, 1 health-estimation, 1 thermal). The criterion now reads: admitted to a cap of 12 and assigned to the function cluster each reviews, with the per-cluster counts stated and matched to the Reviews column of Table 3; the three review-type syntheses (all cross-cutting) remain coded under their dominant method family.

4 — Thermal management no longer called "thin" in the abstract.

The abstract listed thermal management among the "comparatively thin" areas for Tier-1 union shares, but Section 3.3 (and the short version's Section III-B) shows thermal is not scarce at pool level: 99 tagged records, comparable to fault and safety (96), with the subset's small thermal count reflecting the coding scheme and

curation quotas, not field scarcity. The abstract now names only charging control, cell balancing, and vehicle- or fleet-level work as under-served, in both versions.

5 — Inclusion window stated to match the corpus (2020–2027 stamps + one 2019 seed).

Criterion (iv) previously stated "a publication year between 2019 and 2026," which the Crossref-stamped 2027 record violates. The criterion now reads: a Crossref date-stamp of 2020–2027 at retrieval — 798 records dated 2020–2026 plus one early-access record stamped to a February 2027 issue (verified against the Crossref registry; deposited July 2026) and disclosed in Section 3.1 — with a single 2019 exception, the field-defining early-life prediction study, admitted as a forced seed. This matches Table 1's Period row and the abstract's composition statement.

6 — Section 4.1 "only function" claim rescoped.

Section 4.1 said SOC is "the only function where classical estimation theory and machine learning have genuinely fused," contradicting Sections 5.4 and 8.1 (physics-informed and hybrid prognostics; physics-informed structures across functions) and the well-known particle-filter/hybrid RUL literature outside the retrieval. The claim is now subset-scoped and filter-specific: within the classified subset, SOC is where the fusion remains most durable, while the other functions hybridize mainly through physics-model priors; the particle-filter-based hybrid RUL literature is acknowledged as sitting largely outside the capped retrieval, so "only" would claim more than the corpus supports.

7 — Conference-venue check: coverage and window boundaries disclosed.

The check's opening named CDC and IFAC as major control venues but the venue set (ACC, ECC, CCDC, APEC, VPPC, ITEC) covers neither, and the set's window (2019–2025) starts a year earlier than the main pool's 2020–2026. Both boundaries are now stated before the counts: the check does not include CDC or IFAC proceedings, so it probes part of the conference literature rather than censusing it; and of the 14 papers, three are dated 2019 (the venue-scoped queries carried no lower year bound) — excluding them leaves 11 papers, 6 learning-based, with unchanged hardware (2), public-dataset (2), and simulation (1) counts among the learning-based papers; only the not-stated class shrinks from three to one.

8 — Presentation: robustness material moved to a Supplementary Material; abstract restated around three claims.

The abstract was a dense block of numbers and caveats, and the hedging apparatus (Box 1 plus parentheticals throughout) made the paper repeatedly retract its own findings. The interpretation rules, the label-noise scenario derivations, the minimum-detectable-effect analysis, the recall-audit construction, and the two abstract-level probes are now consolidated in a standalone Supplementary Material (S1–S5, built by build_supplement.py and released with the repository), with one-sentence summaries and pointers left in the main text. The abstract was rewritten around three clear, defensible claims — the validation audit (8/88 online, 17/88 leak-prone splits, the two-pole data base), the no-detectable-difference physics-informed contrast (explicitly underpowered, equivalence not claimed), and the within-subset method-family stratification — with scope and robustness stated once in a single closing sentence. "With impunity" (Section 6.2) was replaced with a precise statement of what the plating-aware protocols do. The short version received the parallel arithmetic and thermal fixes but keeps its robustness summary self-contained for the venue's page limits.

Sincerely,
The authors
Leading University, Sylhet, Bangladesh — 18 October 2026